Prefix-Primitive Annihilators of Languages *

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Abstract

This paper studies algebraic properties concerning the p-primitive annihilators of languages. The p-primitive annihilators of languages contains left P_1-annihilators and right P_1-annihilators. There are some interesting results presented in this paper. Such as: for every finite language L, the left P_1-annihilator of L is not equal to the right P_1-annihilator of L, the set $\gamma_{P_1}(L)$ is not regular for any finite language L, and the set $\gamma_{P_1}(L)$ is not empty for any thin language.

1 Introduction

The theory of formal languages plays a considerable role in the field of computer science. Both regular languages and disjunctive languages are especially important applications. Recall that every regular language is accepted by a finite automaton [6]. It is the union of the equivalence classes of a congruence relation of finite index. Moreover, from the definition of disjunctive languages, every disjunctive language has infinitely many congruence classes. This yields that no disjunctive language is regular. For a given language $L \subseteq X^*$, the relation P_L defined on X^* by

$$u \equiv v(P_L) \Leftrightarrow (xuy \in L \Leftrightarrow xvy \in L, \forall x, y \in X^*)$$

is a congruence. A language L is *regular* if P_L is of finite index and is *disjunctive* if P_L is the equality.

A word $f \in X^+$ is a *primitive word* if f is not a power of any other word. It is known that every nonempty word is a power of a primitive word and the expression is unique. The set of all primitive words over X is, by the standard way, denoted by Q. (see [8], [11]) No beginning square words are introduced and studied in [2] and [4]. Since every no beginning square word is primitive, no beginning square words are called prefix primitive words, shortly *p-primitive words*. The set of all p-primitive words over X is denoted by P_1.

The concept of annihilators of languages is proposed by Shyr and Yu in [10]. This is somewhat like the T-conductor of a vector into a given subspace studied in linear algebra. When a binary operation is applied to a given language with another language, we would like to know what family of languages the other language can be chosen from such that the result belongs to a specific family. Let $\mathcal{F}$ be a family of languages and L is a language. Then a language A is called a right $\mathcal{F}$-annihilator of L if LA is a language in $\mathcal{F}$ and a language B is called left $\mathcal{F}$-annihilator of L if BL is a language in $\mathcal{F}$. There are many annihilators had been studied, such as $\mathcal{Q}$-annihilators [10], $\mathcal{D}$-annihilators [12], $\mathcal{P}$-annihilators [12] and d-primitive annihilators [13]. Let $\mathcal{S}$ be the family of all p-primitive languages. Then we denote p-primitive annihilators of languages by $\mathcal{S}$-annihilators. There are many different properties between $\mathcal{S}$-annihilators and $\mathcal{Q}$-annihilators, $\mathcal{D}$-annihilators, $\mathcal{P}$-annihilators. For example, the catenation of languages is $\mathcal{Q}$-commutative but is not $\mathcal{S}$-commutative. This provides us the initial motivation for studying p-primitive annihilators of languages.

2 Definitions and Preliminaries

Let X be a finite alphabet and X^* be the free monoid generated by X. Any element of X^* is called a *word*. For any word $u \in X^*$, let $\lg(u)$ be

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the length of u, which is the number of letters occurring in u. Any subset of X^* is called a *language* over X. Let $X^+ = X^* \setminus \{\lambda\}$, where λ is the empty word. A word $w \in X^+$ is said to be *primitive* if $w = u^n$ with $u \in X^+$ implies $n = 1$. It is known that every nonempty word is a power of a primitive word and the expression is unique. The set of all primitive words over X is denoted by Q. The system $< X^*, \circ >$ is known as a free monoid, where the operation $\circ$ is the catenation of two words. For convenience, for any $u, v \in X^*$, let $u \circ v = uv$. The catenation of two languages A and B is defined to be the set $A \circ B = AB = \{xy \mid x \in A,\ y \in B\}$. For a language L, let $L^{(i)} = \{w^i \mid w \in L\}$ for any $i \geq 1$.

The partial order relation $\leq_{\rm p}$ (resp., $\leq_{\rm s}$) is defined as: for $u, v \in X^*$, $u \leq_{\rm p} v$ (resp., $u \leq_{\rm s} v$) if $v \in uX^*$ (resp., $v \in X^*u$). A word u is said to be a *prefix* (resp., *suffix*) of another word w if $u \leq_{\rm p} w$ (resp., $v \leq_{\rm s} w$). For a given word $u \in X^+$, we define the infix of the word u as $E(u) = \{v \in X^+ | u \in X^*vX^*\}$. The partial order relation $\leq_{\rm d}$ is defined as: for $u, v \in X^*$, $u \leq_{\rm d} v$ if $v = ux = yu$ for some $x, y \in X^*$. Note that $1 \leq_{\rm d} v$ for any $v \in X^+$. For a word $u \in X^+$, let $\mathrm{Pref}(u) = \{v \in X^+ \mid v \leq_{\rm p} u\}$ and $\mathrm{Suff}(u) = \{v \in X^+ \mid v \leq_{\rm s} u\}$. For a language $L \subseteq X^*$, let $L_p = \{v \in X^+ \mid v \leq_{\rm p} u,$ for some $u \in L\}$ and $L_s = \{v \in X^+ \mid v \leq_{\rm s} u,$ for some $u \in L\}$.

A word w is said to be *prefix primitive* (shortly, *p-primitive*), i.e., $w \notin u^2X^*$ for any $u \in X^+$. The set of all p-primitive words over X is denoted by P_1. For example: let $X = \{a, b\}$. Then ab^n is a p-primitive word for any $n \geq 1$. It is clear that P_1 is a subset of Q. A word u is a *conjugate* of another word v if there is a word w such that $uw = wv$. A word $u \in X^+$ is said to be *d-primitive* if for any $v \in X^+$, $v \leq_{\rm d} u$ implies $u = v$. Let $D(1)$ be the set of all d-primitive words in X^* ([3]). It is clear that $D(1)$ is also a subset of Q.

Let $L \subseteq X^+$ be a language and $\mathcal{F}$ be a family of languages. The right $\mathcal{F}$-annihilating function on 2^{X^+}, denoted $\alpha_{\mathcal{F}}$, is defined by $\alpha_{\mathcal{F}}(L) = \{Y \subseteq X^+ \mid LY \in \mathcal{F}\}$ ([12]). Every element in $\alpha_{\mathcal{F}}(L)$ is called a right $\mathcal{F}$-annihilator of the language L. Dually, the left $\mathcal{F}$-annihilating function on 2^{X^+}, denoted $\gamma_{\mathcal{F}}$, is defined by $\gamma_{\mathcal{F}}(L) = \{Y \subseteq X^+ \mid YL \in \mathcal{F}\}$.

3 The P_1-Annihilators of Languages

Some algebraic properties of annihilators concerning p-primitive languages are studied in this section. Let $L \subseteq X^+$ be a language and P_1 be the set of all p-primitive words. The right P_1-annihilating function on 2^{X^+}, denoted α_{P_1}, is defined by $\alpha_{P_1}(L) = \{w \in X^+ \mid Lw \subseteq P_1\}$. Every element in $\alpha_{P_1}(L)$ is called a right P_1-annihilator of the language L. Dually, the left P_1-annihilating function on 2^{X^+}, denoted γ_{P_1}, is defined by $\gamma_{P_1}(L) = \{w \in X^+ \mid wL \subseteq P_1\}$. We give some examples of annihilators of p-primitive languages.

Example 3.1 Let X be an alphabet. Then

(1) $\alpha_{P_1}(X) = \emptyset$, and $\alpha_{P_1}(P_1) = \emptyset$.

(2) For any $a \in X$, $\alpha_{P_1}(\{a\}) = (X \setminus \{a\})X^* \setminus \{xaxy \mid x \in X^+, y \in X^*\}$.

(3) For any $a \in X$, $\gamma_{P_1}(\{a\}) = P_1 \setminus \{xax \mid x \in X^+\}$.

From the definitions of right P_1-annihilators and left P_1-annihilators, the following results are clear.

Lemma 3.1 ([1]) *Let $L \subseteq X^+$. The following statements are true:*

(1) If $\alpha_{P_1}(L) \neq \emptyset$, then $L \subseteq P_1$.

(2) If $\gamma_{P_1}(L) \neq \emptyset$, then $\gamma_{P_1}(L) \subseteq P_1$.

Lemma 3.2 ([1]) *Let $L \subseteq X^+$. Then if $u \in \alpha_{P_1}(L)$, then $v \in \alpha_{P_1}(L)$, for every $v \leq_{\rm p} u$.*

Proposition 3.1 *Let $L \subseteq X^+$. The following statements are true:*

(1) If $\alpha_{P_1}(L) = \emptyset$, then $\alpha_{P_1}(A) = \emptyset$, for every language $A \supseteq L$.

(2) If $\gamma_{P_1}(L) = \emptyset$, then $\gamma_{P_1}(B) = \emptyset$, for every language $B \supseteq L$.

Proof. We only show that Statement (1) is true and the proof of Statement (2) is similar. Since $\alpha_{P_1}(L) = \emptyset$, we have that $Lu \not\subseteq P_1$ for every $u \in X^*$. That is, for every $u \in X^*$, there exists a word $w \in L$ such that $wu \notin P_1$. This implies that for every $u \in X^*$ there exists $w \in L \subseteq A$ such that $wu \notin P_1$. Hence $Au \not\subseteq P_1$. That is, $\alpha_{P_1}(L) = \emptyset$. #

From Example 3.1, we see that $\alpha_{P_1}(\{a\}) \neq \gamma_{P_1}(\{a\})$. That is, $\alpha_{P_1}(L)$ need not equal to $\gamma_{P_1}(L)$, for any $L \subseteq X^*$. We now give some languages L such that $\alpha_{P_1}(L) = \gamma_{P_1}(L)$.

Lemma 3.3 *Let $a \neq b \in X$ and let $L = \{ba^{\lg(u)+1}uba^{\lg(u)+1} \mid u \in X^+ \setminus \{a\}\}$. Then $\alpha_{P_1}(L) = \gamma_{P_1}(L) = \{a\}$. Indeed, by the definition of $\lg(u)$, we have that $ba^{\lg(u)+1}uba^{\lg(u)+1} \in P_1$. That is, $L \subseteq P_1$. For any $u \in X^+ \setminus \{a\}$, it is clear that $Lu = \emptyset$ and $La \subseteq P_1$. That is, $\alpha_{P_1}(L) = \{a\}$. On the other hand, it is clear that $aL \subseteq P_1$ and $uL = \emptyset$ for any $u \in X^+ \setminus \{a\}$. That is, $\gamma_{P_1}(L) = \{a\}$. Thus $\alpha_{P_1}(L) = \gamma_{P_1}(L) = \{a\}$.*

Example 3.2 Let $a \neq b \in X$ and let $L = \{ba^{\lg(u)+1}uba^{\lg(u)+1} \mid u \in X^+ \setminus \{a, ab\}\}$. Then $\alpha_{P_1}(L) = \gamma_{P_1}(L) = \{a, ab\}$. Indeed, by the definition of $\lg(u)$, we have that $ba^{\lg(u)+1}uba^{\lg(u)+1} \in P_1$. That is, $L \subseteq P_1$. For any $u \in X^+ \setminus \{a, ab\}$, it is clear that $Lu = \emptyset$ and $Lab, La \subseteq P_1$. That is, $\alpha_{P_1}(L) = \{a, ab\}$. On the other hand, it is clear that $aL, abL \subseteq P_1$ and $uL = \emptyset$ for any $u \in X^+ \setminus \{a, ab\}$. That is, $\gamma_{P_1}(L) = \{a, ab\}$. Thus $\alpha_{P_1}(L) = \gamma_{P_1}(L) = \{a, ab\}$.

A language $L \subseteq X^*$ is *dense* if for any $w \in X^*$, there exist $x, y \in X^*$ such that $xwy \in L$. That is, for every $w \in X^*$, $X^*wX^* \cap L \neq \emptyset$. A language is called *thin* if it is not dense. In the rest of this section, we show that every thin language L, the left P_1-annihilator $\gamma_{P_1}(L)$ is not empty. First, we give some known lemmata we need. From the definitions of α_{P_1} and γ_{P_1}, it is clear that $\alpha_{P_1}(X) = \emptyset$ and $\gamma_{P_1}(X^*) = \emptyset$. Let $L \subseteq X^+$. In Proposition 3.2, we give characteristics for that $\alpha_{P_1}(L) \neq \emptyset$ and $\gamma_{P_1}(L) \neq \emptyset$.

Lemma 3.4 ([2]) *Let u, v be two distinct words with $u \in P_1(X)$. Then uv is a p-primitive word if and only if none of the following statements holds:*

(1) $u <_{\mathrm{p}} v$,

(2) $u = x_1x_2x_1$ with $x_2 \leq_{\mathrm{p}} v$ where $x_1, x_2 \in X^+$, or

(3) $v = x_1ux_1y$ where $x_1 \in X^+$, $y \in X^*$.

Proposition 3.2 ([1]) *Let $L \subseteq X^+$. The following statements are true:*

(1) $\alpha_{P_1}(L) \neq \emptyset$ if and only if $L \subseteq P_1$ and $La \subseteq Q$ for some $a \in X$.

(2) $\gamma_{P_1}(L) \neq \emptyset$ if and only if there exists a p-primitive word, say $u \in P_1$, such that both of the following conditions hold:

(i) $xux \notin L_p$ for every $x \in X^*$, and

(ii) $u \notin yL_py$ for every $y \in X^+$.

From above proposition, the following remark is trivial:

Remark 3.1 *Let $L \subseteq X^+$. The following statements are true:*

(1) $\alpha_{P_1}(L) = \emptyset$ if and only if one of the following conditions holds: (i) $L \nsubseteq P_1$, (ii) $L \subseteq P_1$ and $La \nsubseteq Q$ for every $a \in X$.

(2) $\gamma_{P_1}(L) = \emptyset$ if and only if, for every $u \in P_1$, one of the following conditions holds: (i) $xux \in L_p$ for some $x \in X^*$, (ii) $u \in yL_py$ for some $y \in X^+$.

(3) If $L \subseteq P_1$ and $La \subseteq Q$ for some $a \in X$, then $La \subseteq P_1$ and then $a \in \alpha_{P_1}(L)$. Moreover, $xax \notin L$ for every $x \in X^*$.

(4) If $u \in P_1$ such that both of (i) $xux \notin L_p$ for every $x \in X^*$, and (ii) $u \notin yL_py$ for every $y \in X^+$ are true, then $u \in \gamma_{P_1}(L)$. Moreover, $zuzv \notin L$ for every $z, v \in X^*$.

Proposition 3.3 *Let $L \subseteq X^+$. If L is thin, then $\gamma_{P_1}(L) \neq \emptyset$.*

Proof. Suppose that L is thin. Then there exists a word $w \in X^+$ such that $L \cap X^*wX^* = .$ Let $a \neq b \in X$ and let $u = ba^{\lg(w)}wba^{\lg(w)}$. It is clear that $u \in P_1$. We claim that $u \in \gamma_{P_1}(L)$. On the other contrary, assume that $u \notin \gamma_{P_1}(L)$. Then there exists $v \in L$ such that $uv \notin P_1$. This in conjunction with $u \in P_1$, there exists $z \in X^+$ with $z \leq_{\mathrm{p}} v$ such that $uz = f^2$, where $f \in Q$. That is, $ba^{\lg(w)}wba^{\lg(w)}z = f^2$. This implies that $f = ba^{\lg(w)}w$ or $ba^{\lg(w)}wba^{\lg(w)} \leq_{\mathrm{p}} f$. If $f = ba^{\lg(w)}w$, then $z = w$. This implies that $w = z \leq_{\mathrm{p}} v$ and then $w \in L_p$. This contradicts to $L \cap X^*wX^*$. If $ba^{\lg(w)}wba^{\lg(w)} \leq_{\mathrm{p}} f$, then there exists $x \in X^*$, $y \in X^+$ with $v = xy$ such that $f = ba^{\lg(w)}wba^{\lg(w)}x = y$. This implies $xba^{\lg(w)}wba^{\lg(w)}x = xy = v \in L$. Thus $L \cap X^*wX^* \neq \emptyset$, a contradiction. Hence $u = ba^{\lg(w)}wba^{\lg(w)} \in \gamma_{P_1}(L)$. #

A language is a 2-code if for every two distinct words in the language constructs a code. ([11]) That is, $L \subseteq X^+$ is a 2-code if for every $u, v \in L$, $\{u, v\}$ is a code.

Proposition 3.4 ([1]) *Let $L \subseteq X^+$ be a language. If $\gamma_{P_1}(L) \neq \emptyset$, then $\gamma_{P_1}(L)$ is a 2-code.*

For the case right P_1-annihilators, the above propositions are not true. Indeed, if $L = X$, then $\alpha_{P_1}(L) = \emptyset$ and hence Propositions 3.3 and 3.4 are not true.

4 The P_1-Annihilators of finite languages

In this section we consider some algebraic properties of p-primitive annihilators for finite languages. We also investigate the relation between left P_1-annihilators and right P_1-annihilators for finite languages. First we give a known result.

Proposition 4.1 ([1]) *Let $L \subseteq X^+$ be a finite language. Then $\gamma_{P_1}(L)$ is dense. Hence $\gamma_{P_1}(L)$ is not an empty set.*

In the following, we show that $\gamma_{P_1}(L)$ is not regular for any finite languages. Note that a language L is *regular* if P_L is of finite index and is *disjunctive* if P_L is the equality. The union of two disjunctive languages is not necessarily disjunctive. It is known that Q is disjunctive and $Q^{(i)}$ is regular for any $i \geq 2$.

From the definitions of disjunctivity and regularity it is clear that a regular language can never be disjunctive and also a disjunctive language can never be regular. To show that the left p-primitive annihilators of finite languages is not regular, we need the following lemma.

Lemma 4.1 ([7]) *Let $L \subseteq X^*$. Then L is disjunctive if and only if for any two words u and v with the same length, $u \equiv v(P_L)$ implies $u = v$.*

The concept of quasi-disjunctive pairs ([3]), ([9]) is useful for showing the disjunctivities of languages. For two disjoint nonempty languages $L_1, L_2 \subseteq X^*$, the pair (L_1, L_2) is said to be a *quasi-disjunctive pair* if for any two distinct words $u, v \in X^*$ with $\lg(u) = \lg(v)$, there exist $x, y \in X^*$ such that $xuy \in L_1$ and $xvy \notin L_2$, or vice versa. From Lemma 4.1, it is clear that if (L_1, L_2) is a quasi-disjunctive pair, then L_1 and L_2 are disjunctive.

Proposition 4.2 *Let $L \subseteq X^+$ be a finite language. Then $\gamma_{P_1}(L)$ is disjunctive. That is, $\gamma_{P_1}(L)$ is not regular.*

Proof. Let $L \subseteq X^+$ be a finite language. Then there exists a positive number $m \geq 1$ such that $m = max\{\lg(z) | z \in L\}$. To prove our result is to show that $(\gamma_{P_1}(L), X^+ \setminus \gamma_{P_1}(L))$ is a quasi-disjunctive pair. Assume that $u \neq v \in X^n$ for any $n \geq 1$. Let $x = ba^{m+n+1}, y = wba^{m+n+1}u$ for some $w \in L$ with $\lg(w) = m$. Then $xuy = ba^{m+n+1}uwba^{m+n+1}u$ and $xvy = ba^{m+n+1}vwba^{m+n+1}u$. From $xuy = ba^{m+n+1}uwba^{m+n+1}u$, since $w \in L$, we have that $xuyw = (ba^{m+n+1}uw)^2$. Thus $xuyL \not\subseteq P_1$, that is, $xuy \notin \gamma_{P_1}(L)$. On the other hand, from $xvy = ba^{m+n+1}vwba^{m+n+1}u$, we have that $xvyL \subseteq P_1$, that is, $xvy \in \gamma_{P_1}(L)$. By Lemma 4.1, $(\gamma_{P_1}(L), X^+ \setminus \gamma_{P_1}(L))$ is a quasi-disjunctive pair. Thus $\gamma_{P_1}(L)$ is disjunctive, hence $\gamma_{P_1}(L)$ is not regular. #

The result is not true for right annihilator. That is, there exists a finite language $L \subseteq X^+$ such that $\alpha_{P_1}(L)$ is finite. We give some examples as follows. Let $X = \{a, b\}$.

Example 4.1 Let $L = \{ab, ba, ba^2b, ab^3a\}$. Then it is clear that $a, b \in \alpha_{P_1}(L)$. But $a^2, ab, ba, b^2 \notin \alpha_{P_1}(L)$. Since $X^+ = \{a, b\} \cup \{a^2, ab, ba, b^2\}X^*$, we have that $\alpha_{P_1}(L) = \{a, b\}$ is finite.

Example 4.2 Let $L = \{ba^3b, ba^2b^2, ba^2baba, bab^3, aba^3, ab^2abab, ab^2a^2, ab^3a\}$. Then it is clear that $a^2, ab, ba, b^2 \in \alpha_{P_1}(L)$ and by Lemma 3.2, $a, b \in \alpha_{P_1}(L)$. But $a^3, a^2b, aba, ab^2, ba^2, bab, b^2a, b^3 \notin \alpha_{P_1}(L)$. Since $X^+ = X \cup X^2 \cup X^3X^*$, we have that $\alpha_{P_1}(L) = \{a, b, a^2, ab, ba, b^2\}$ is finite.

Proposition 4.3 *Let L be a finite language. Then $\alpha_{P_1}(L) \neq \gamma_{P_1}(L)$.*

Proof. Let L be a finite language. By Proposition 4.1, $\gamma_{P_1}(L)$ is dense. If $\alpha_{P_1}(L) = \emptyset$, then it is clear that $\alpha_{P_1}(L) \neq \gamma_{P_1}(L)$. Now suppose that $\alpha_{P_1}(L) \neq \emptyset$. Then by Remark 3.1, there exists $a \in X$ such that $La \subseteq P_1$, that is, $a \in \alpha_{P_1}(L)$. Assume that $\alpha_{P_1}(L) = \gamma_{P_1}(L)$. This implies that $a \in \gamma_{P_1}(L)$ and then $a \notin L_p$. Hence there exists another letter b such that $b \in L_p$. Thus $b \notin \gamma_{P_1}(L)$. By assumption that $\alpha_{P_1}(L) = \gamma_{P_1}(L)$, we have that $b \notin \alpha_{P_1}(L)$. On the other hand, since L is finite, there exists $m \geq 1$ such that $m = max\{\lg(w) \mid w \in L\}$. Hence $ba^m u \in P_1$ for every $u \in L$. Thus $ba^m \in \gamma_{P_1}(L)$. This in conjunction with $\alpha_{P_1}(L) = \gamma_{P_1}(L)$, we get that $ba^m \in \alpha_{P_1}(L)$. But by Lemma 3.2, $b \in \alpha_{P_1}(L)$. This contradicts to that $b \notin \alpha_{P_1}(L)$. Hence $\alpha_{P_1}(L) \neq \gamma_{P_1}(L)$ for any finite language L. #

5 Conclusion

This paper studies algebraic properties concerning the p-primitive annihilators of languages. The p-primitive annihilators of languages contains left P_1-annihilators and right P_1-annihilators. We show that for every finite language L, the left P_1-annihilator of L is not equal to the right P_1-annihilator of L. We also show that the set $\gamma_{P_1}(L)$ is not regular for any finite language L. Moreover, the set $\gamma_{P_1}(L)$ is not empty for any thin language. In future work, Languages L such that $\gamma_{P_1}(L) = \alpha_{P_1}(L)$ will be investigated related algebraic properties.

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